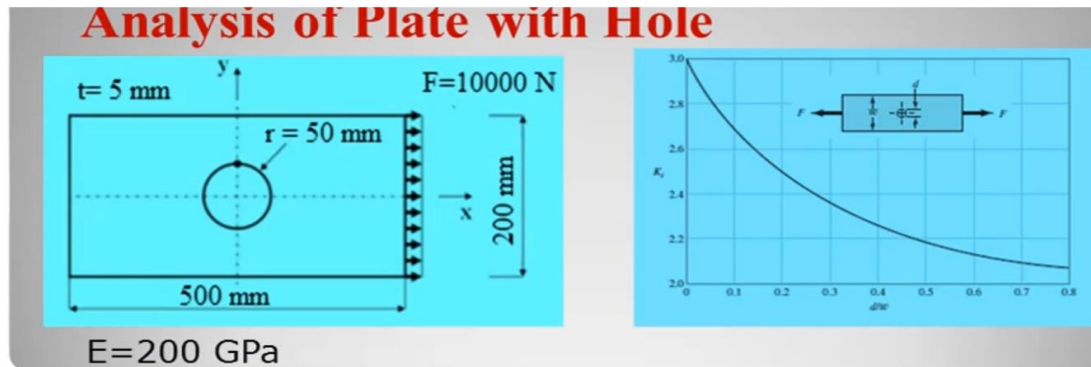


Practical 3

Title – Plate Element – Structural Linear Analysis

Sample Example



SOLUTION:

$$\sigma_{\max} = \sigma_{\text{avg}} \times K_c$$

$$\begin{aligned}\sigma_{\text{avg}} &= F/A \\ &= \frac{F}{(b-d)t} = \frac{10000}{(200-100)5} \\ &= 20 \text{ N/mm}^2\end{aligned}$$

$$d/b = 100/200 = 0.5$$

$$\therefore K_c \approx 2.15$$

$$\begin{aligned}\sigma_{\max} &= \sigma_{\text{avg}} \times K_c \\ &= 20 \times 2.15\end{aligned}$$

$$\sigma_{\max} = 43 \text{ N/mm}^2$$

Ac
Go

Procedure

Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

Step 2 Pre-processor

Pre-processor – Element Type – Add – Solid – Quad 4 node 182 - OK - Close

Options – Plane stress with thickness – OK- Close

Real Constants – Add – Add – Thickness 5 mm - Ok

Step 3 Material

Material Properties – Material Models – Structural – Linear – Elastic –

Isotropic – Enter Young's Modulus – Enter Poisson's Ratio - OK

Step 4 Modeling

Modeling – Create – Areas – Rectangle – By 2 corners - width 250, Height 100 – OK

Modeling – Create – Areas – Circle – Solid Circle – Radius 50 – OK

**Modeling – Operate – Booleans – Subtract – Areas – 1st area Rectangle – OK
– 2nd Area Circle - OK**

Step 5 Meshing

**Meshing – Mesh Tools – Smart Size - Fine – Mesh – Select Rectangle – Mesh
– Click Raise Hidden tool - Close**

Boundary Conditions

**Loads – Define Loads – Apply – Structural – Displacement – Symmetry
Boundary Condition – On Lines – select 1st and 2nd lines – OK**

Step 6 Apply Loads

**Loads – Define Loads –Apply – Structural – Pressure (Convert Load into
Pressure) – On Lines – Select Line – OK – enter pressure value - OK**

Step 7 Solution

Solution – Solve – Current LS – OK

Step 8 General Results

**General Post Process – Plot Results Contour Plot – Nodal Solution – DOF
Solution – Displacement Vector Sum - OK**

To Find Stresses

Apply Command – Set, last

**General Post Process – Plot Results Contour Plot – Nodal Solution – Stresses
– Von Mises Stresses - OK**

To See Actual Plate

**Plot Controls – Style – Symmetry Expansion – Periodic Cycle – select $\frac{1}{4}$
symmetry - OK**

To See Thickness of Plate

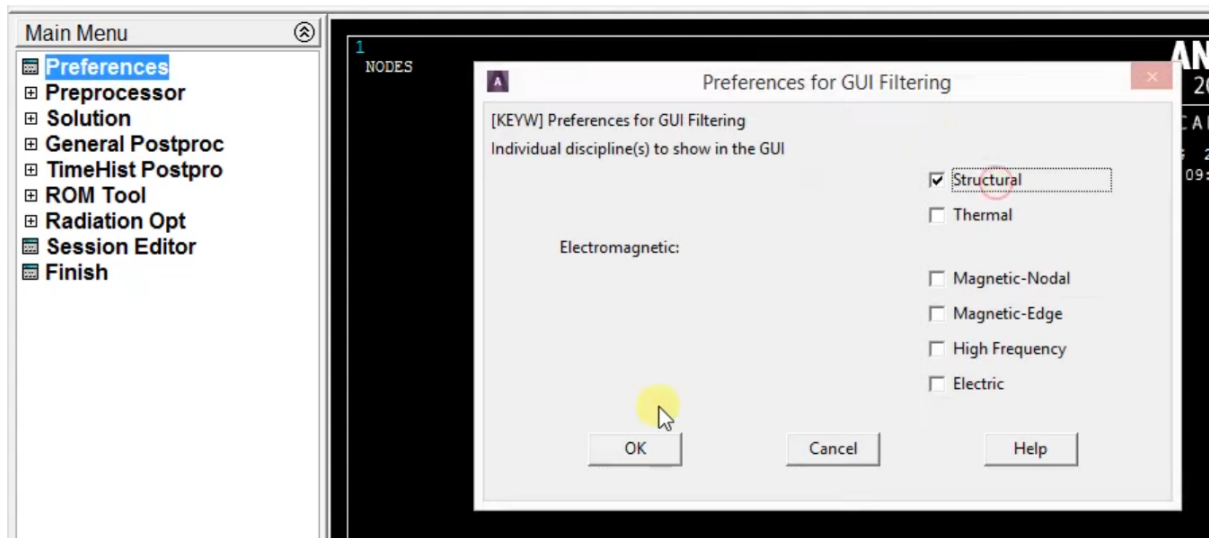
Plot Controls – Style – Size and Shape – Display of Element on – OK

To See Animation

**Plot Control – Animate – Deformed Results – stresses – von mises stresses -
ok**

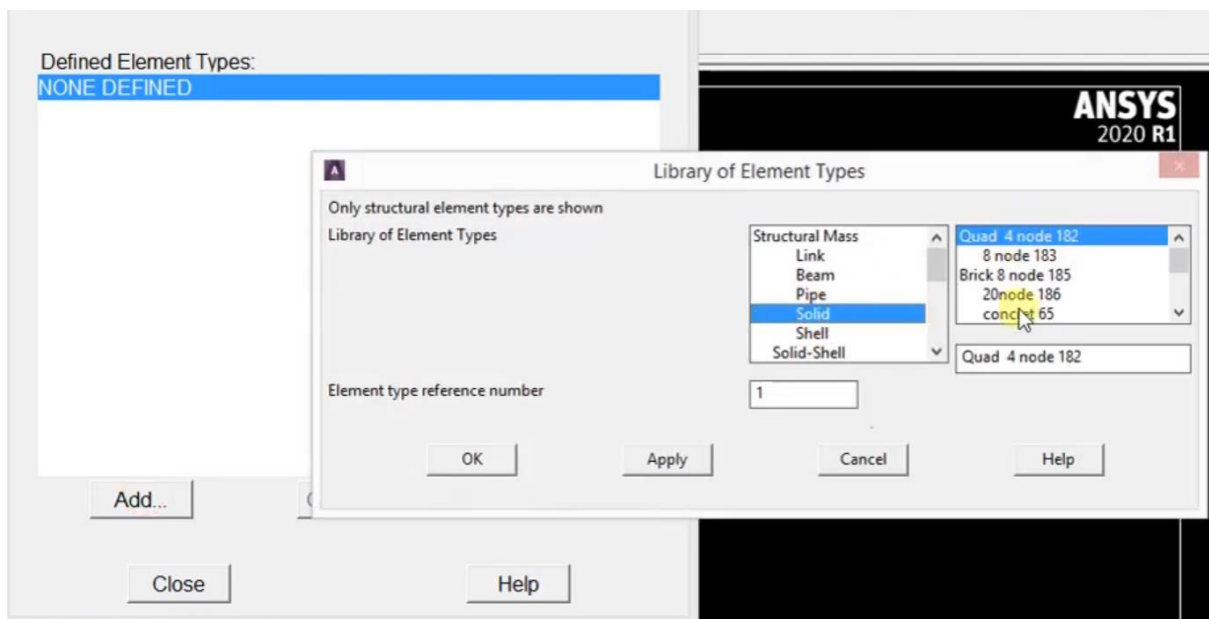
Step 1 Preferences

Open Ansys APDL – Preferences – Structural – OK

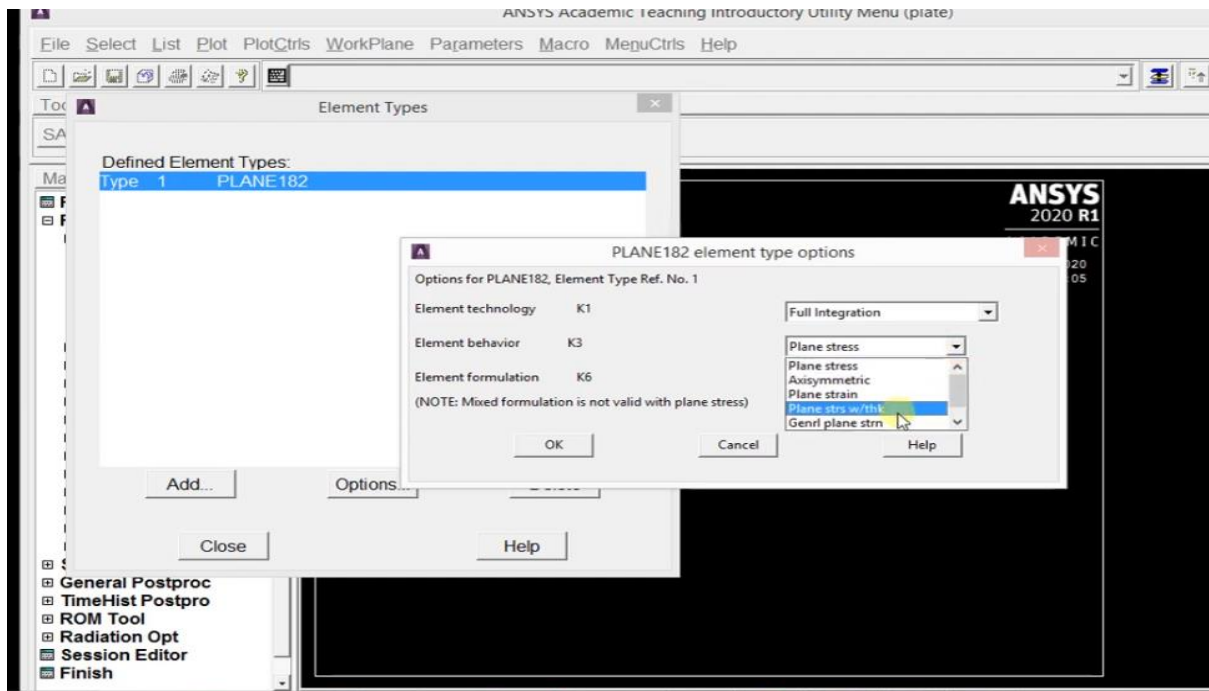


Step 2 Pre-processor

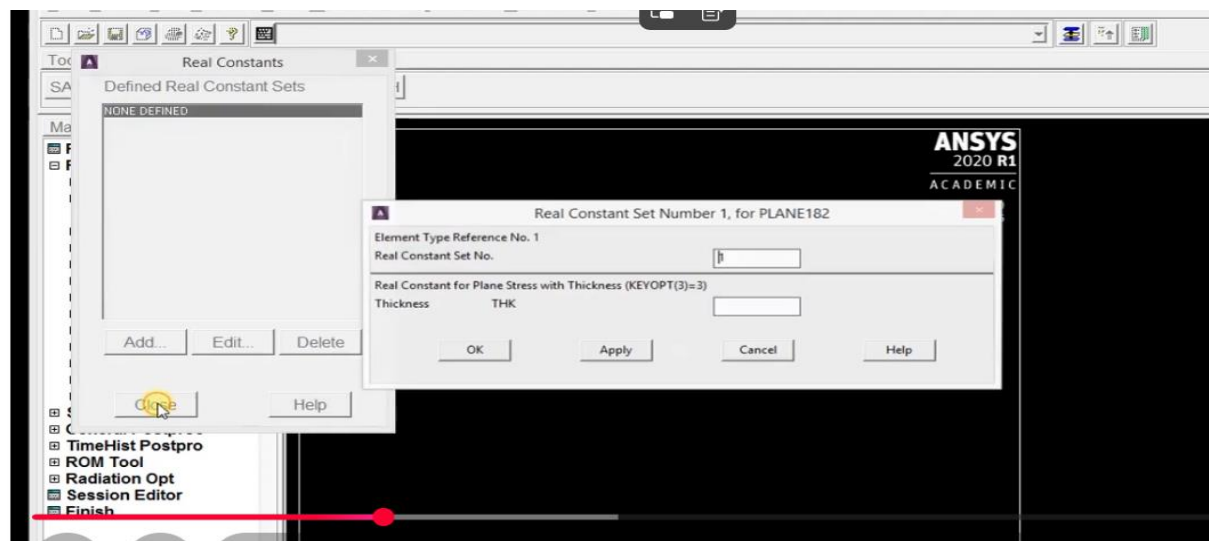
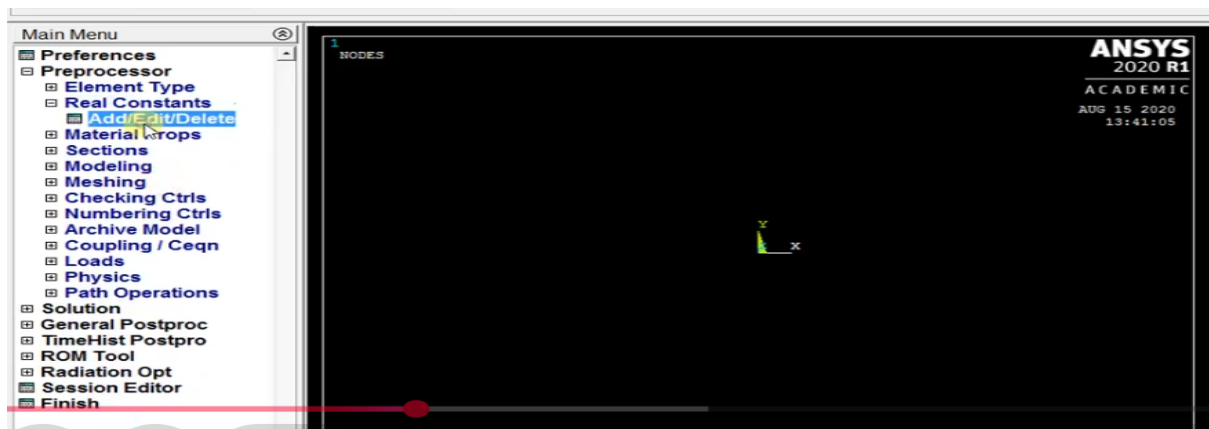
Pre-processor – Element Type – Add – Solid – Quad 4 node 182 - OK - Close

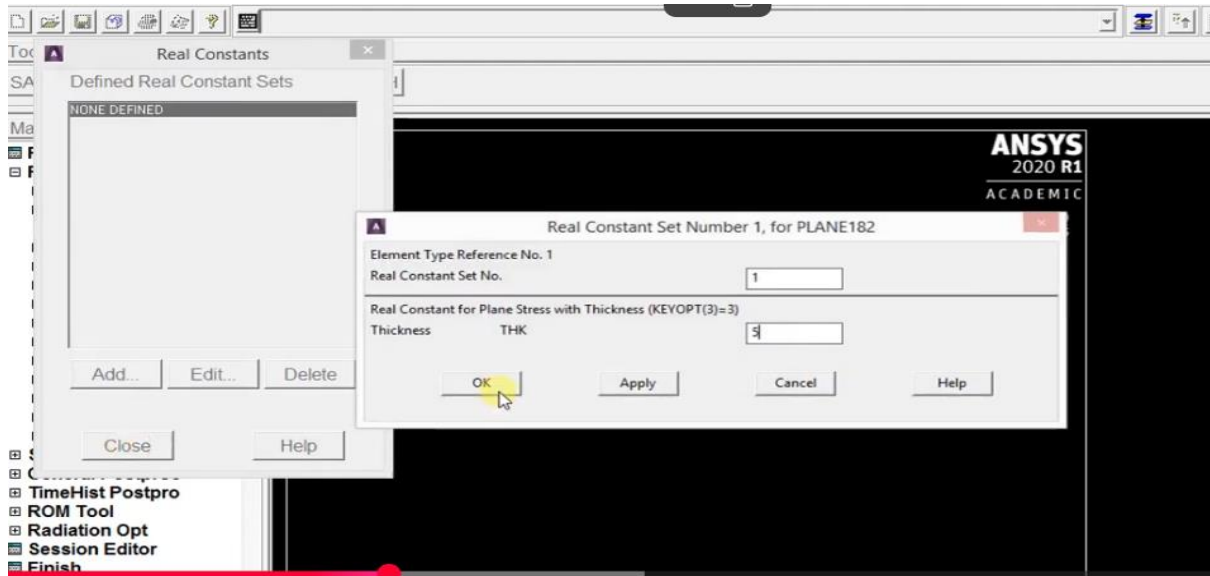


Options – Plane stress with thickness – OK- Close



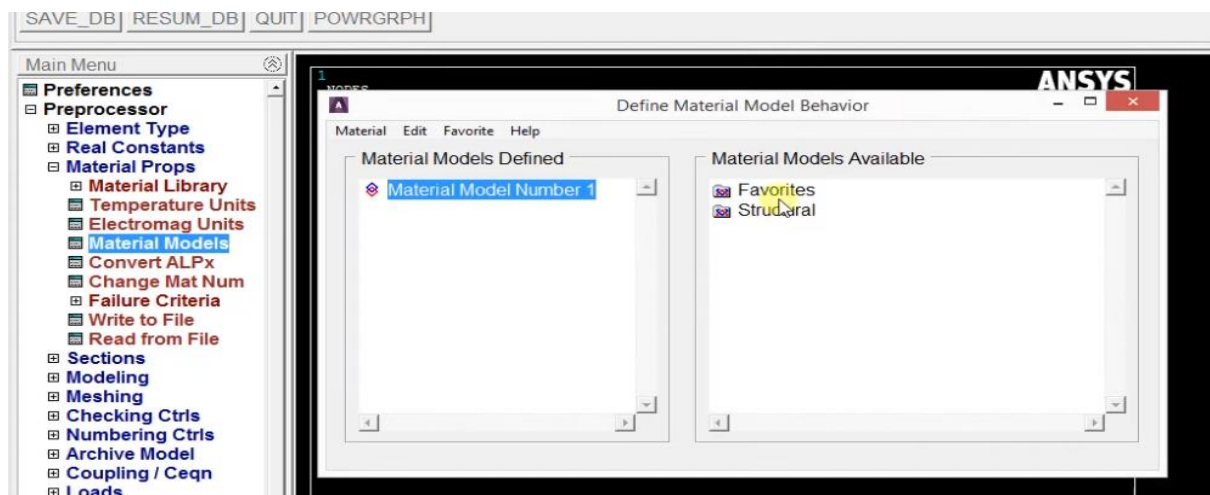
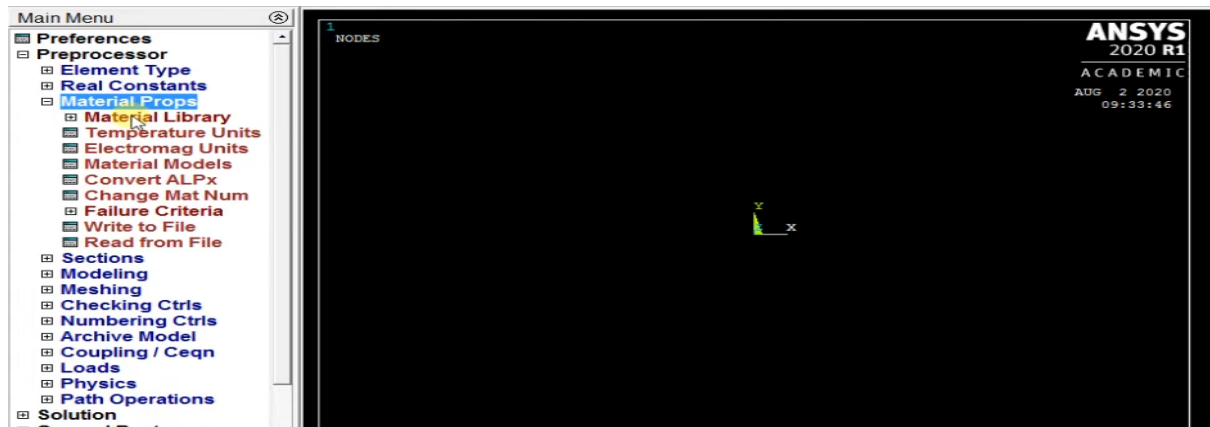
Real Constants – Add – Add – Thickness 5 mm - Ok

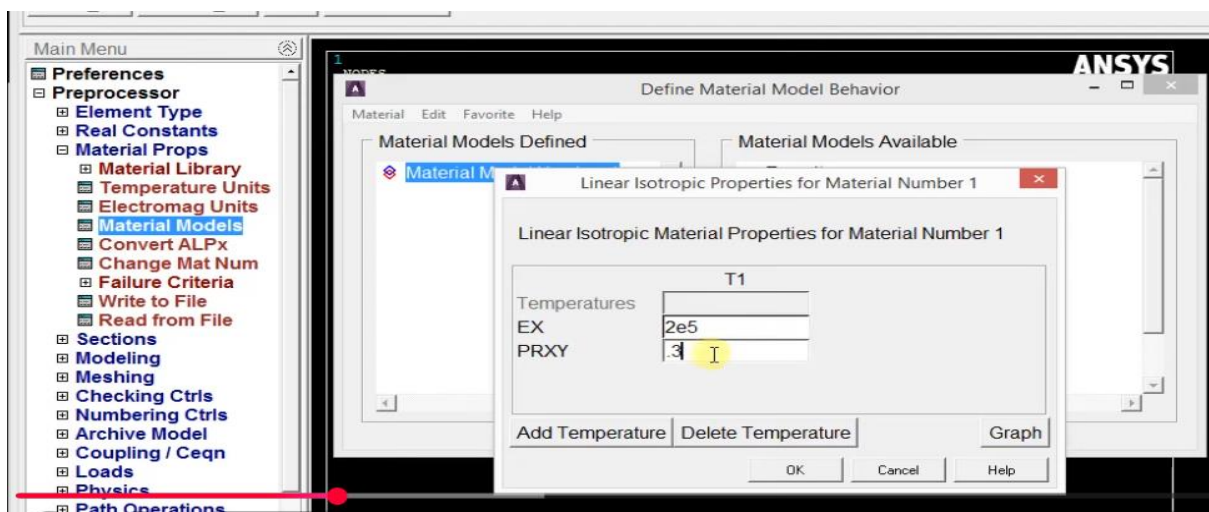
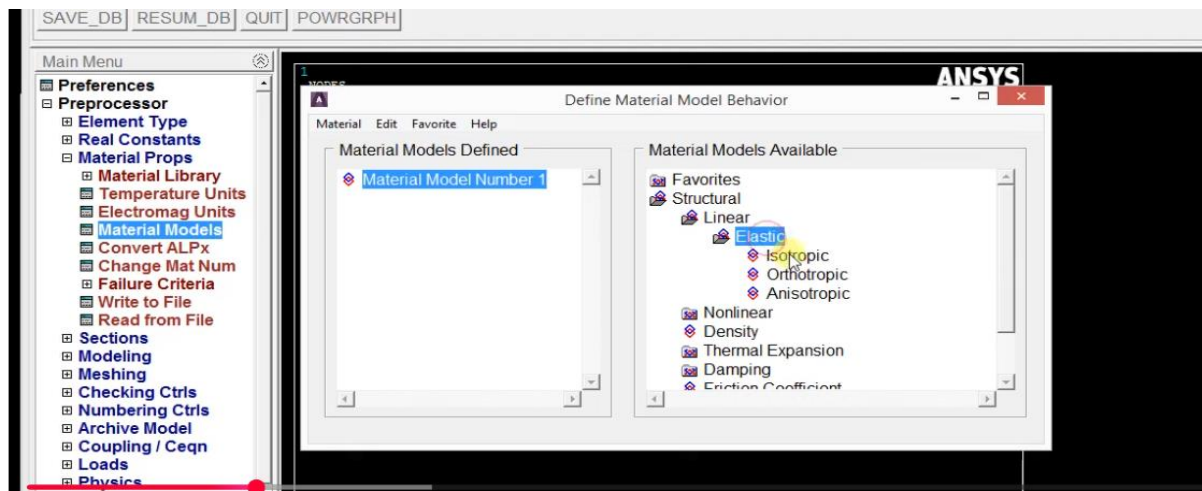




Step 3 Material

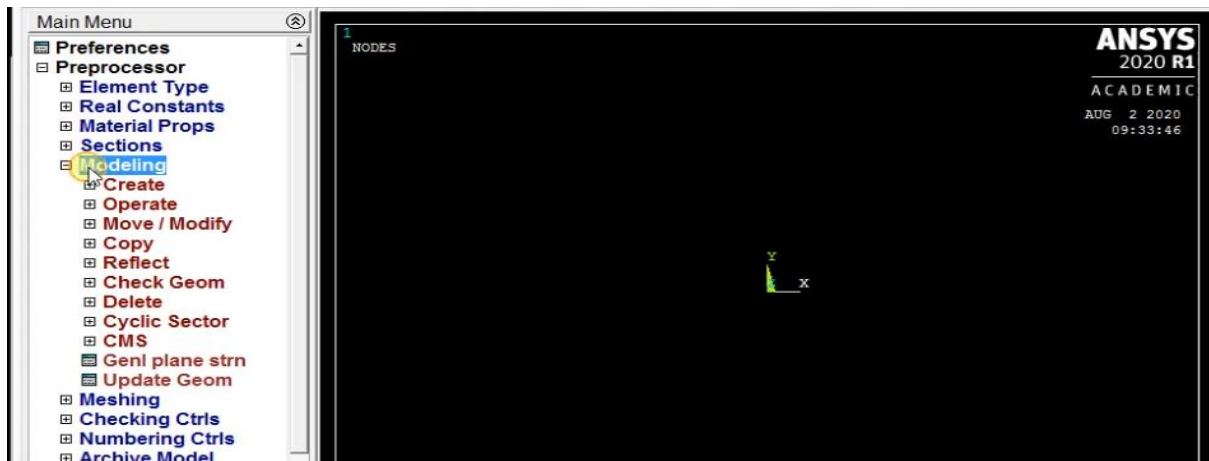
Material Properties – Material Models – Structural – Linear – Elastic – Isotropic – Enter Young’s Modulus – Enter Poisson’s Ratio - OK

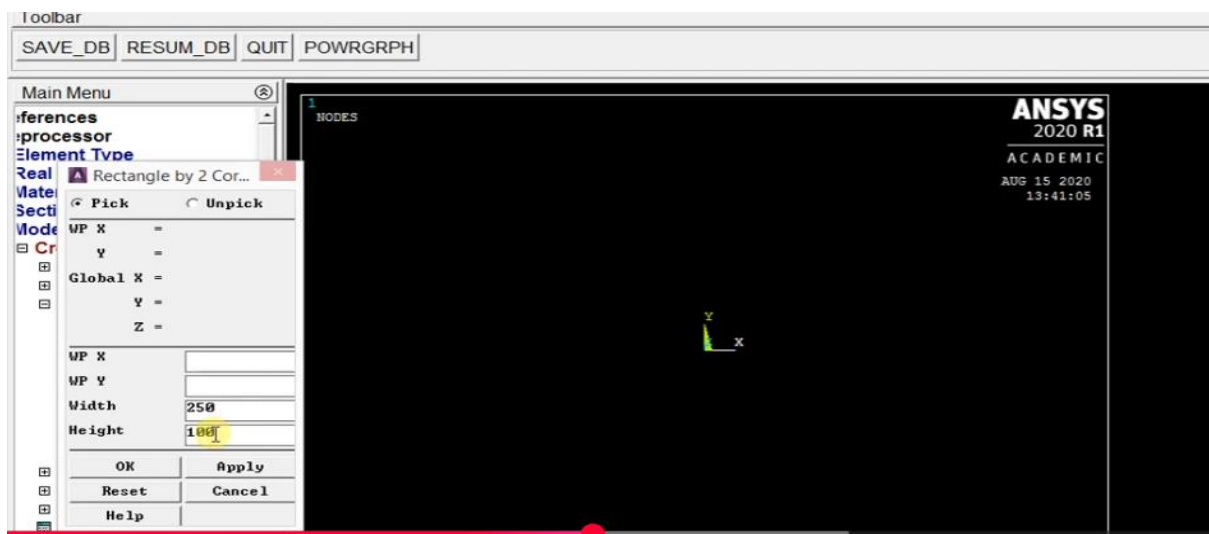
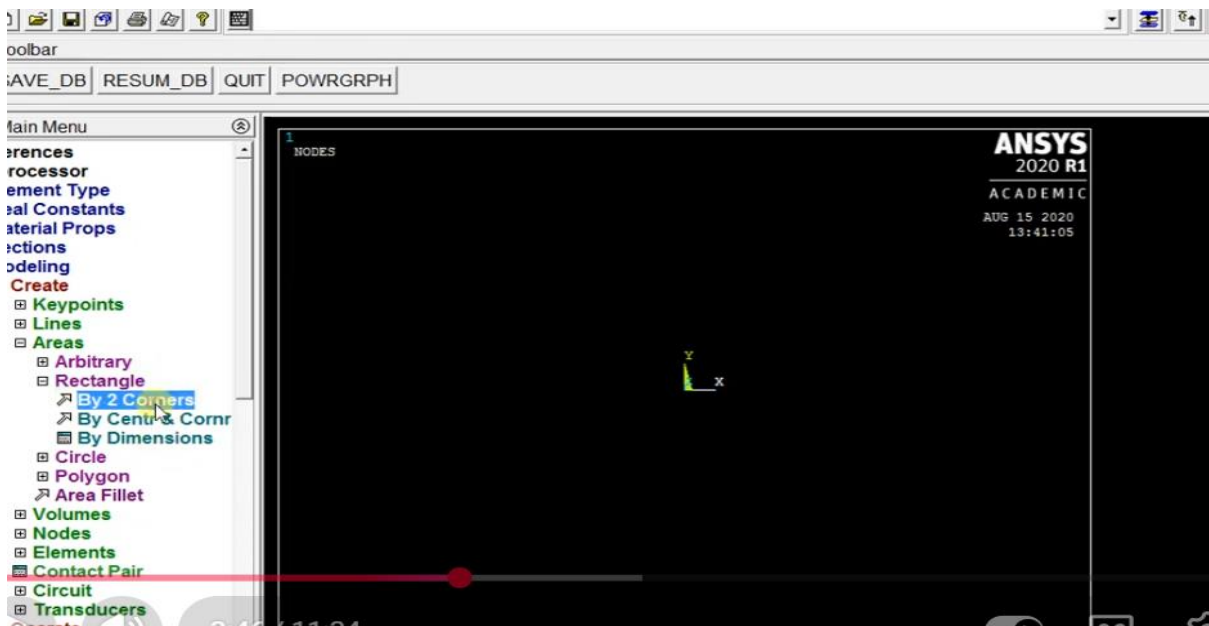
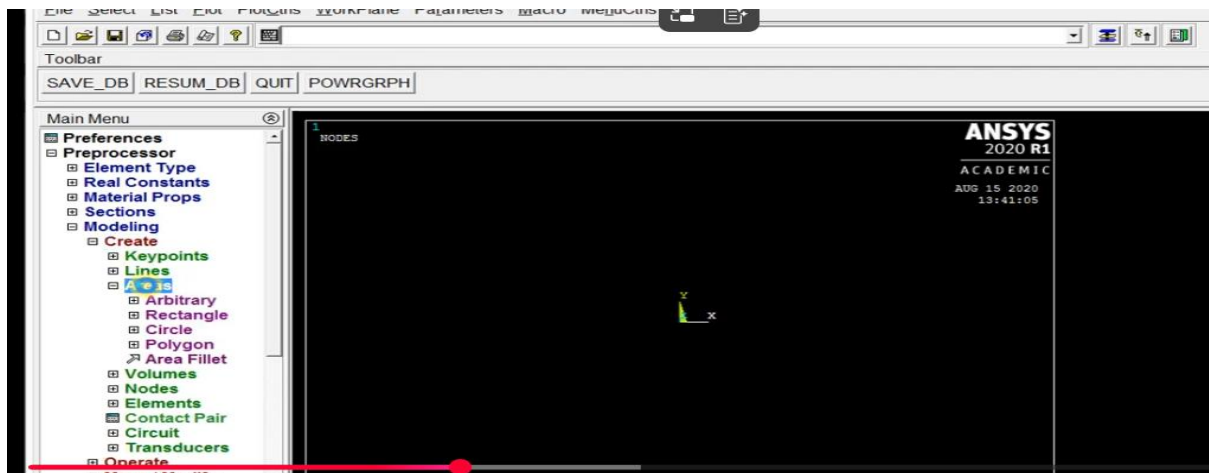




Step 4 Modeling

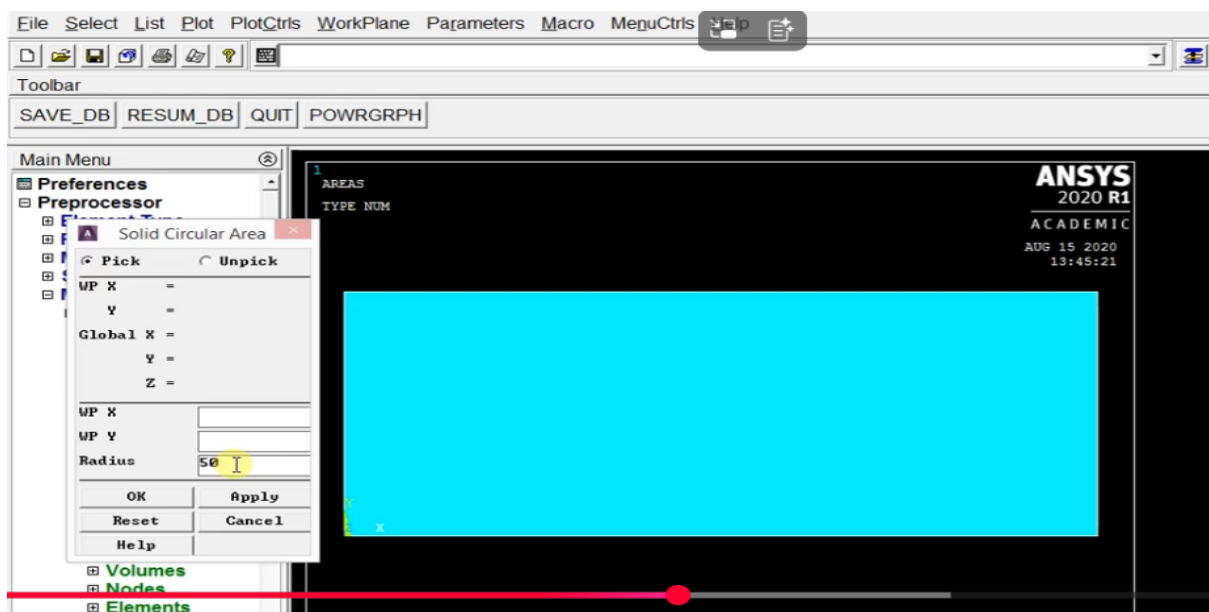
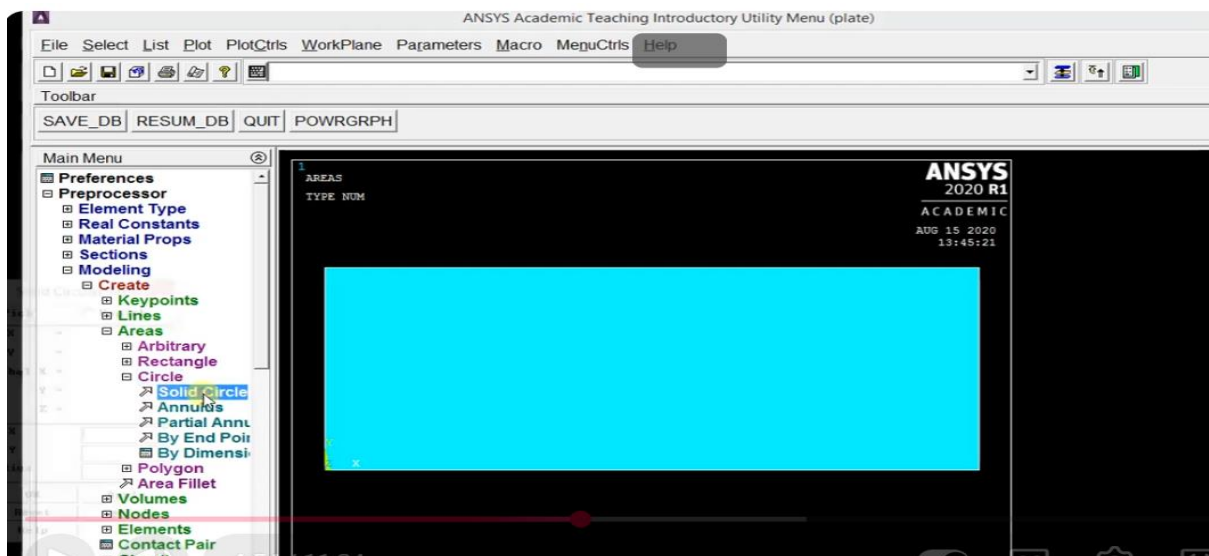
Modeling – Create – Areas – Rectangle – By 2 corners - width 250, Height 100 - OK

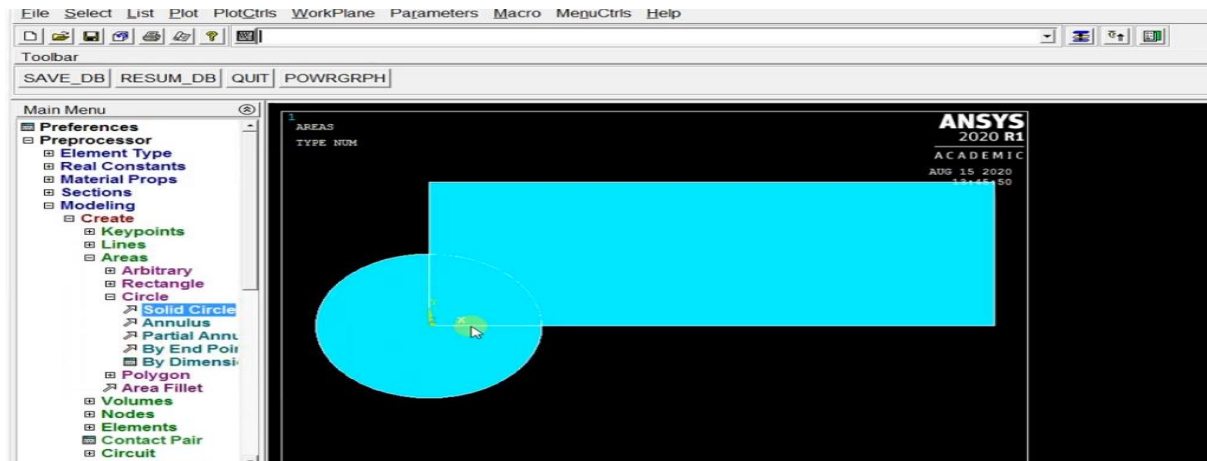




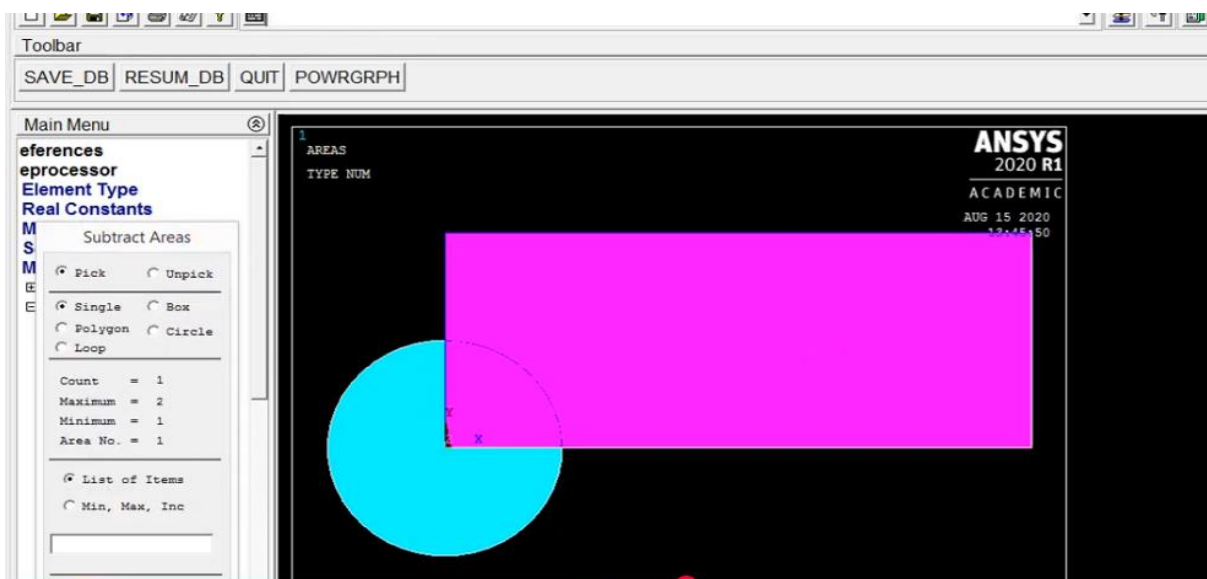
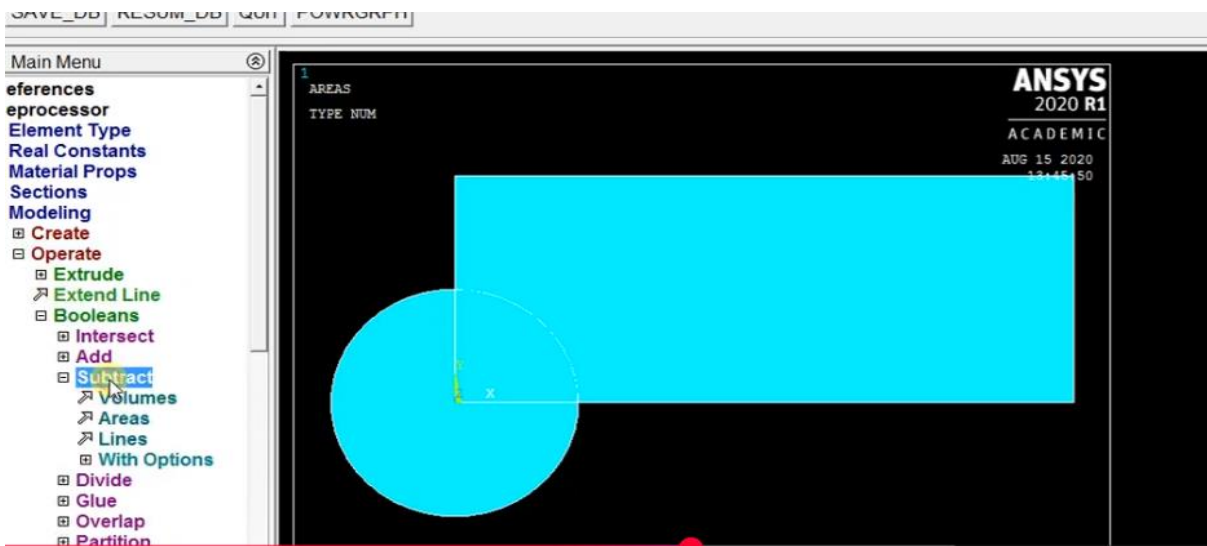


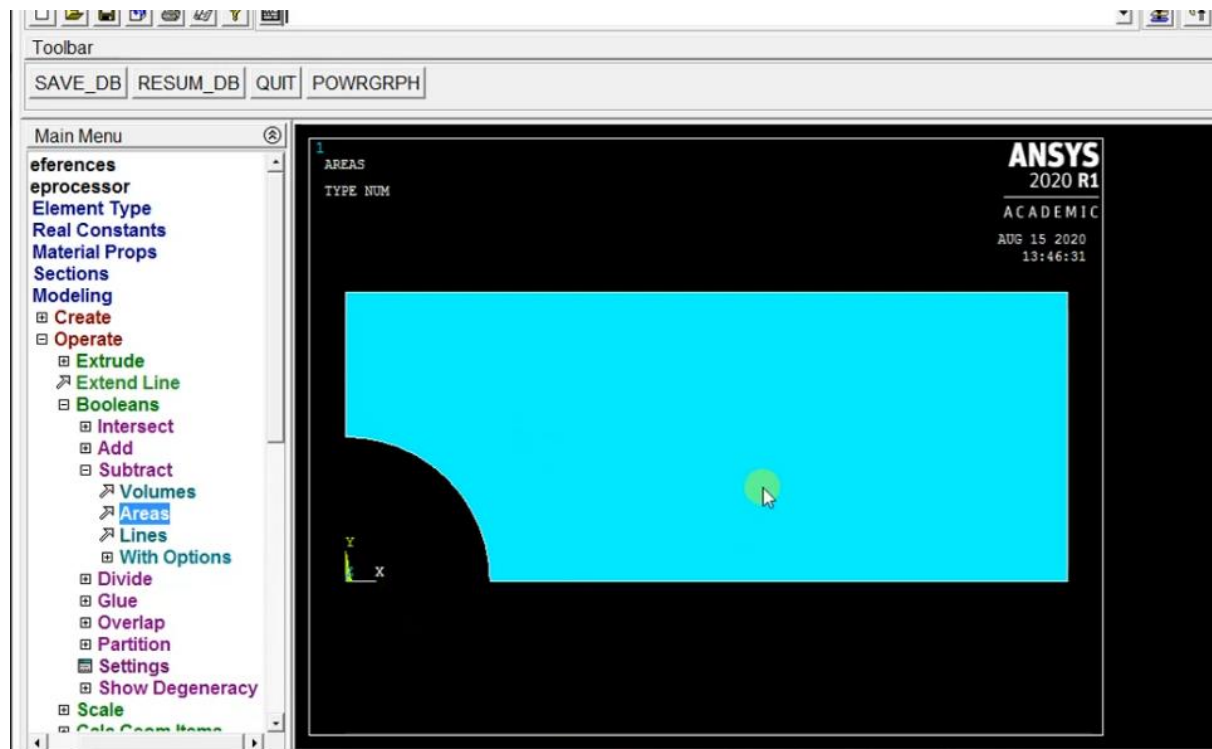
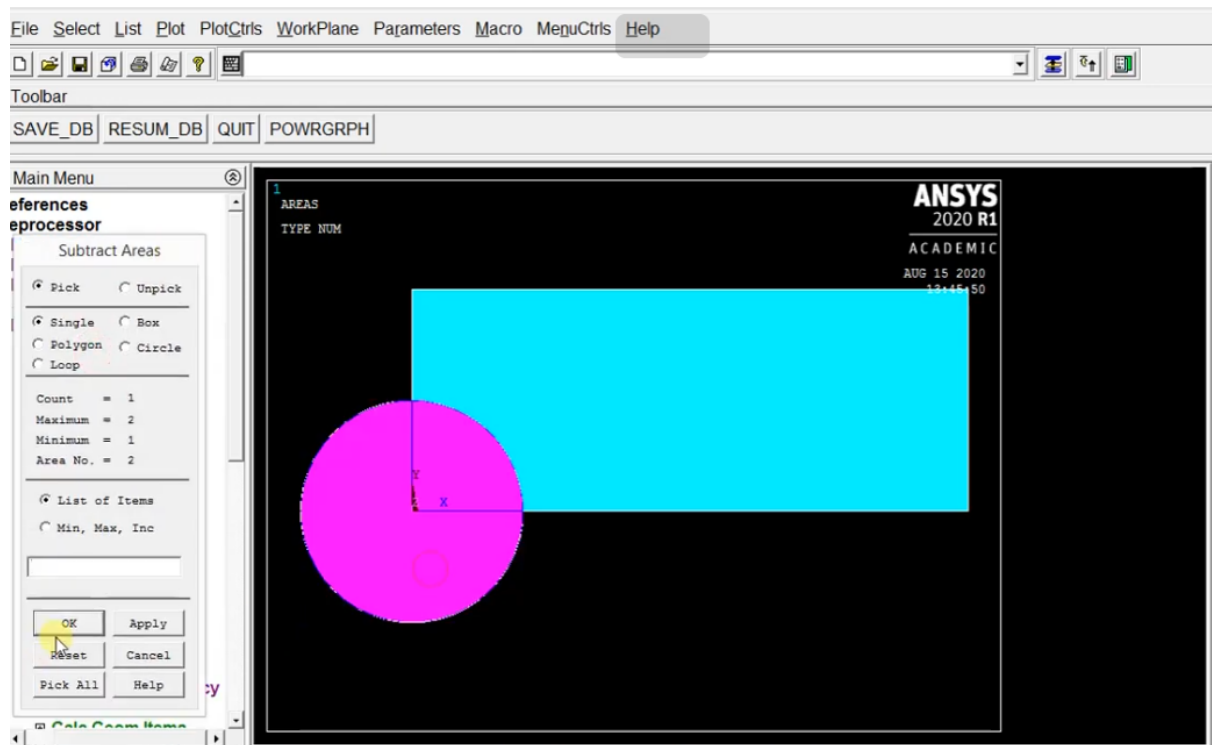
Modeling – Create – Areas – Circle – Solid Circle – Radius 50 - OK





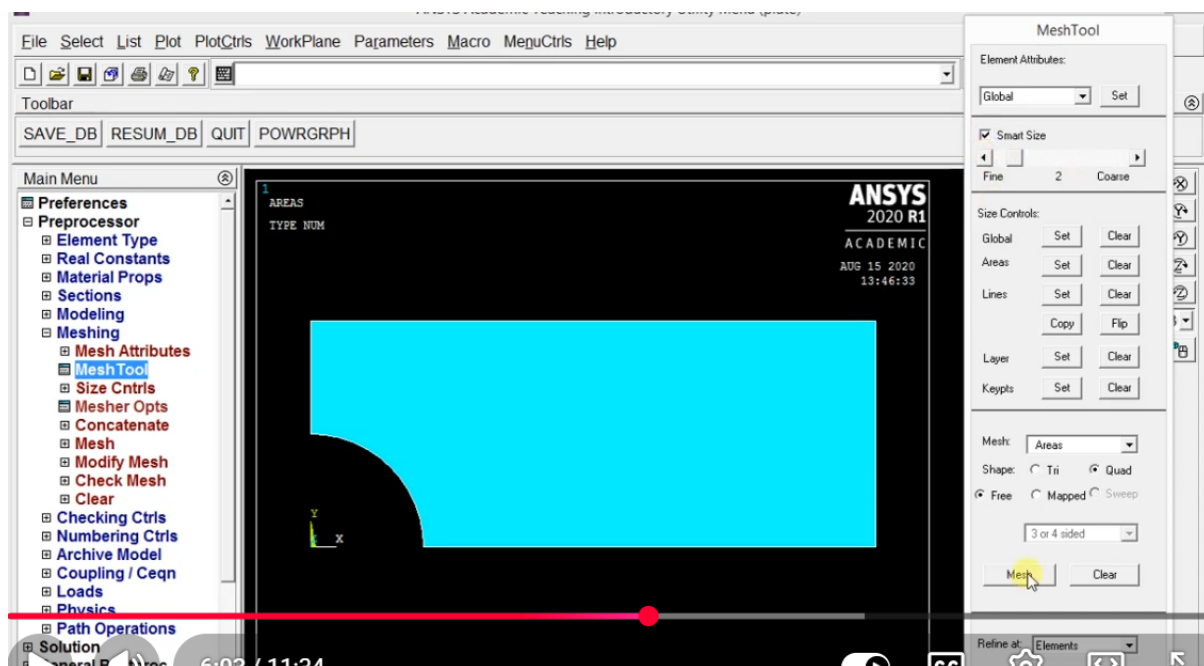
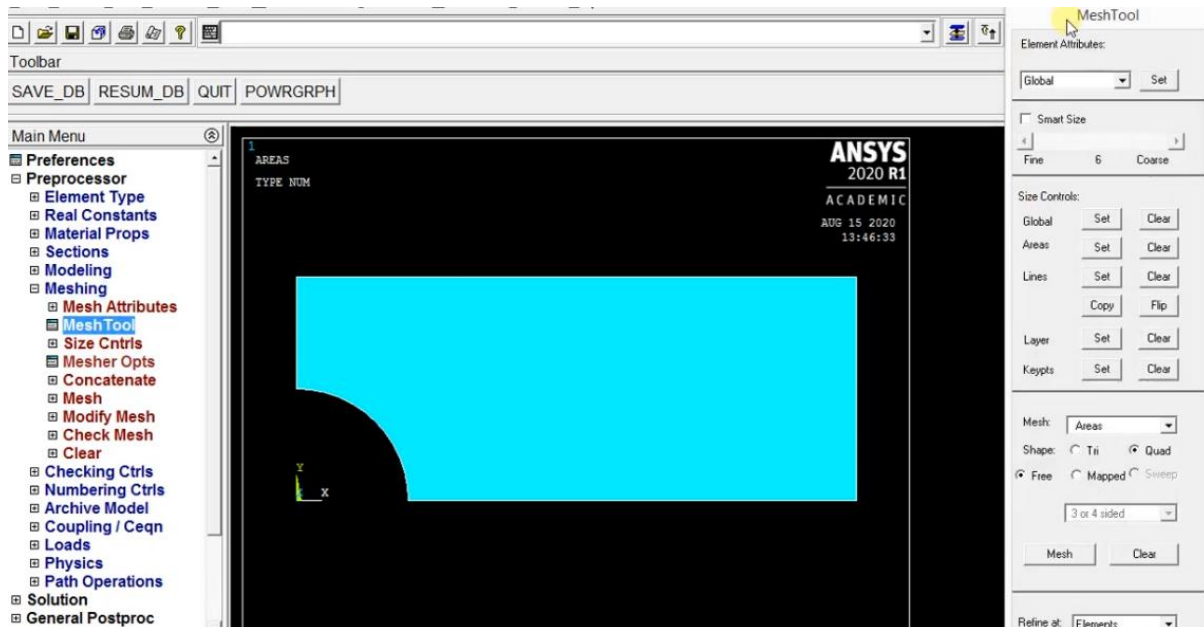
Modeling – Operate – Booleans – Subtract – Areas – 1st area Rectangle – OK
– 2nd Area Circle - OK

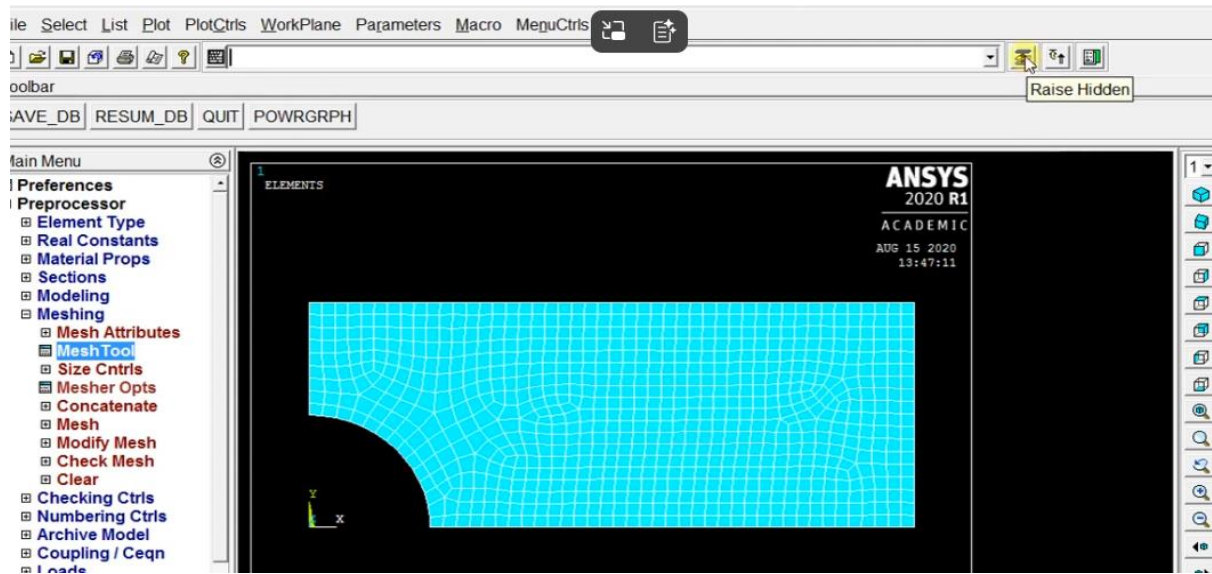
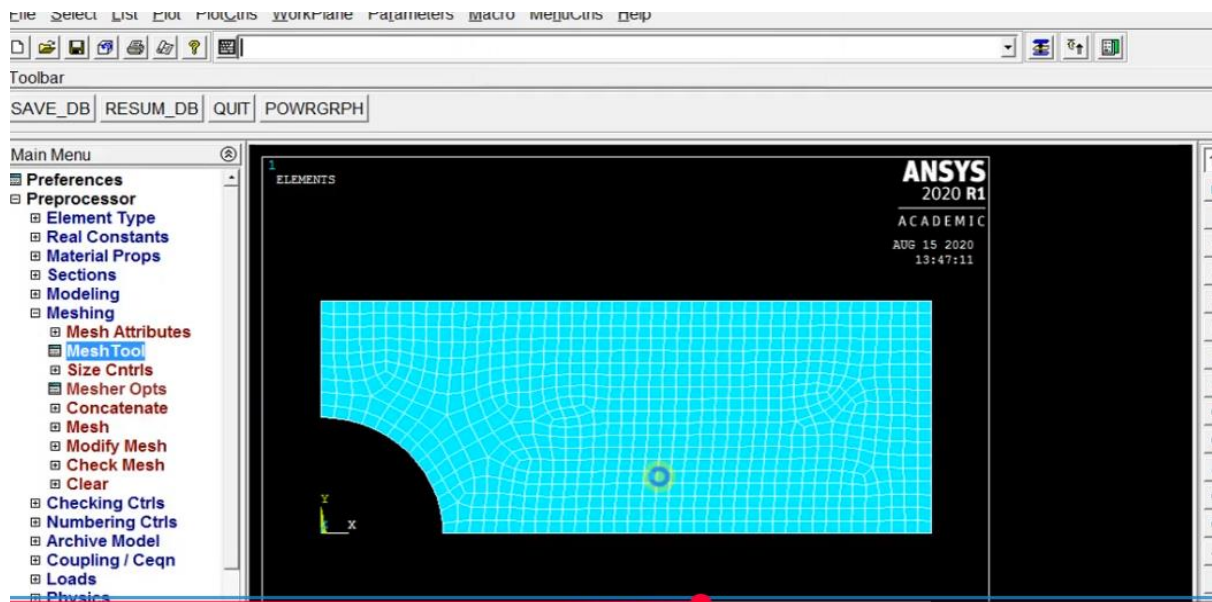
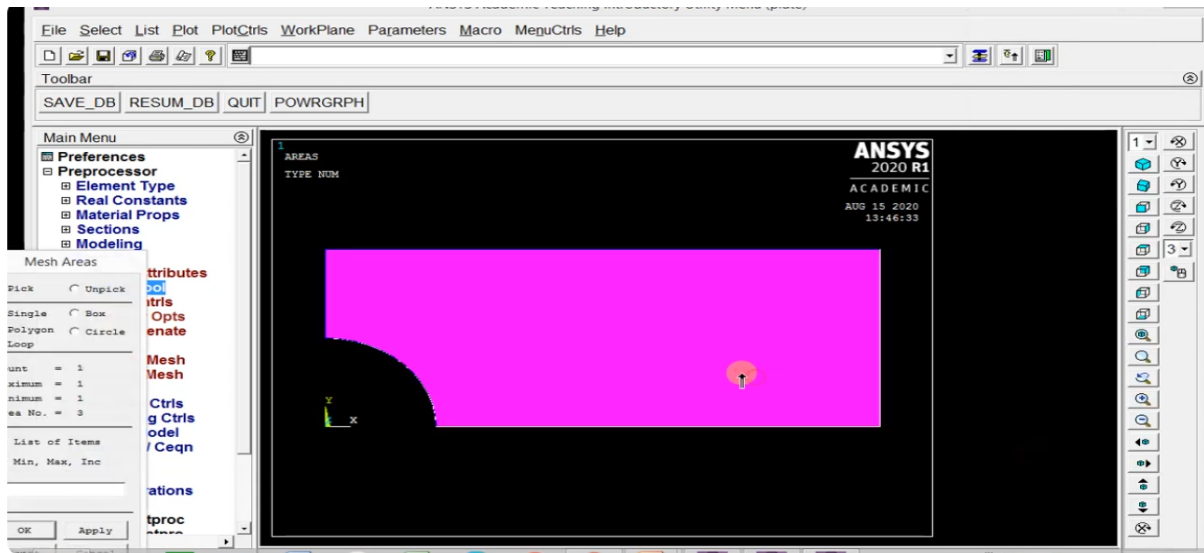




Step 5 Meshing

**Meshing – Mesh Tools – Smart Size - Fine – Mesh – Select Rectangle – Mesh
– Click Raise Hidden tool - Close**

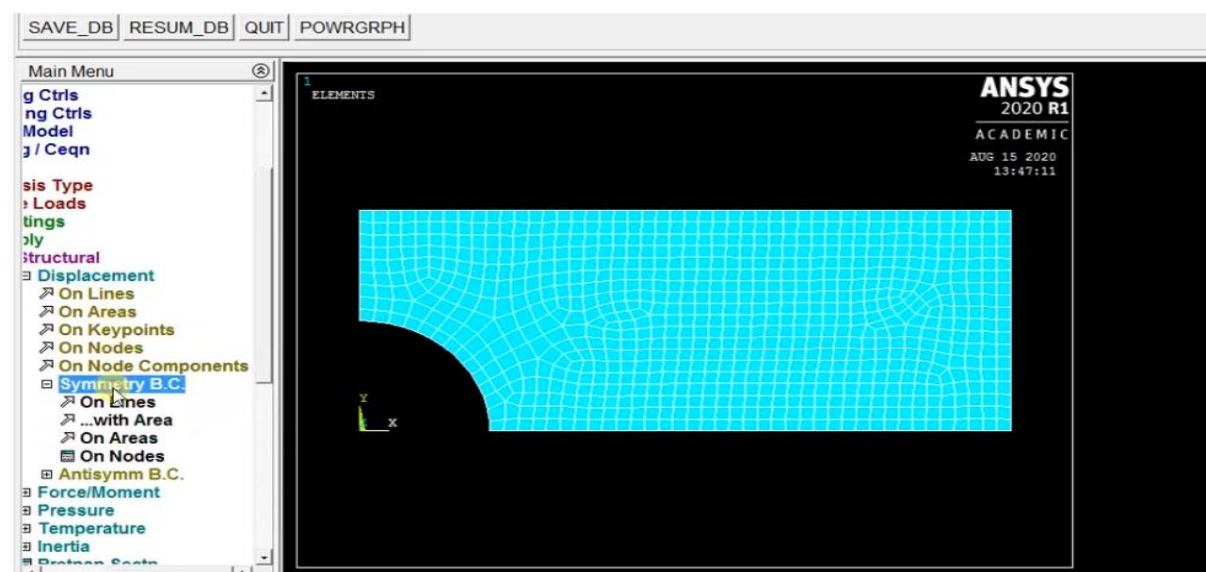
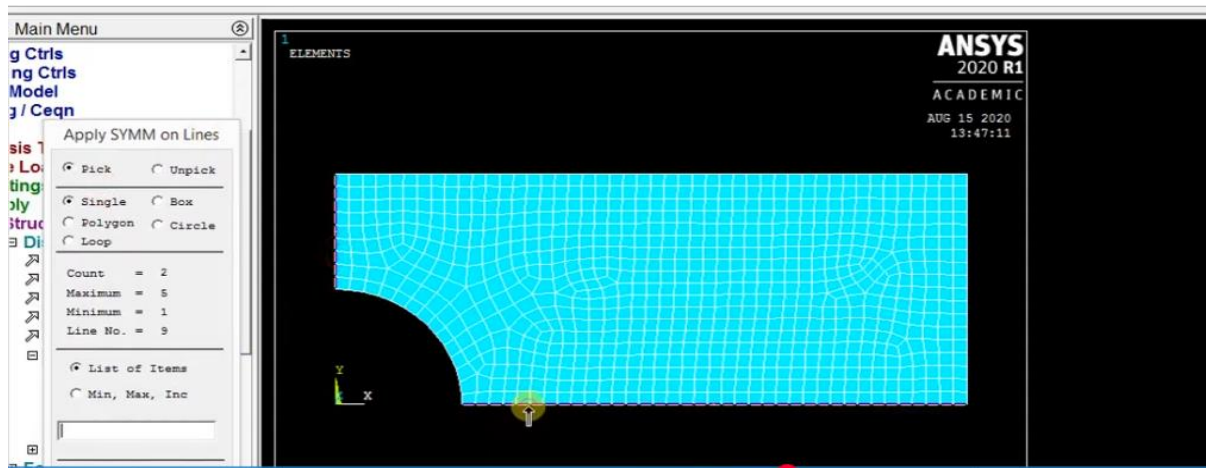
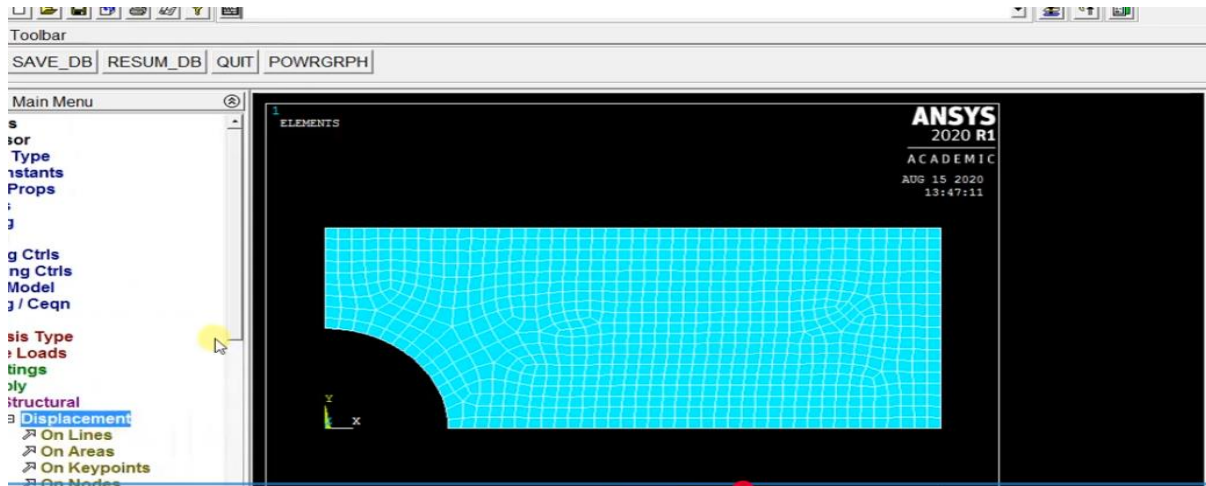


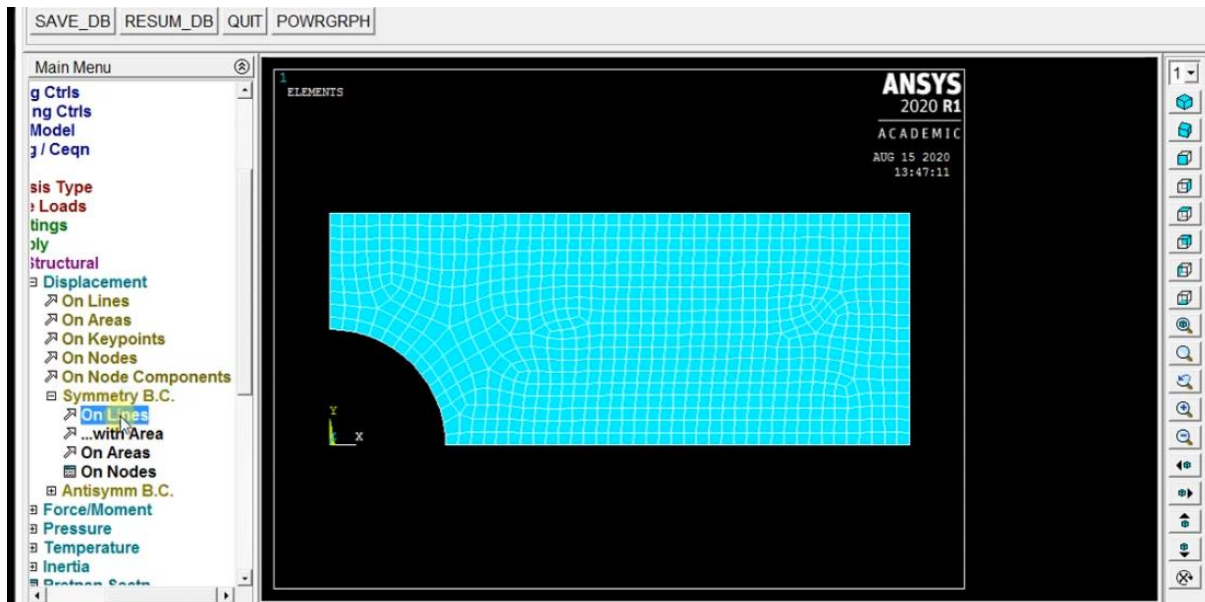


Boundary Conditions

Loads – Define Loads – Apply – Structural – Displacement – Symmetry

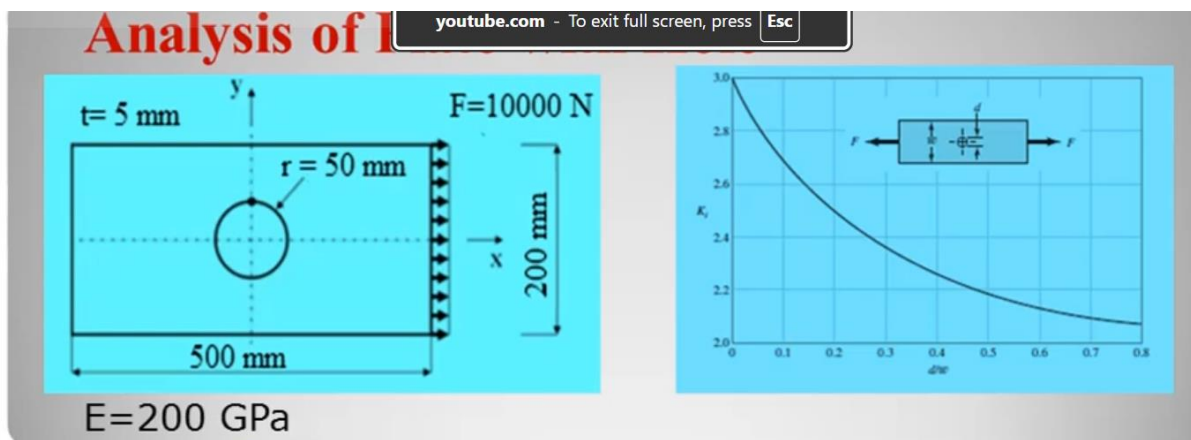
Boundary Condition – On Lines – select 1st and 2nd lines - OK





Step 6 Apply Loads

Loads – Define Loads –Apply – Structural – Pressure (Convert Load into Pressure) – On Lines – Select Line – OK – enter pressure value - OK



SOLUTION:

Convert given force into pressure
(parameter in ANSYS)

$$P = F/A = \frac{10000}{200 \times 5} = 10$$

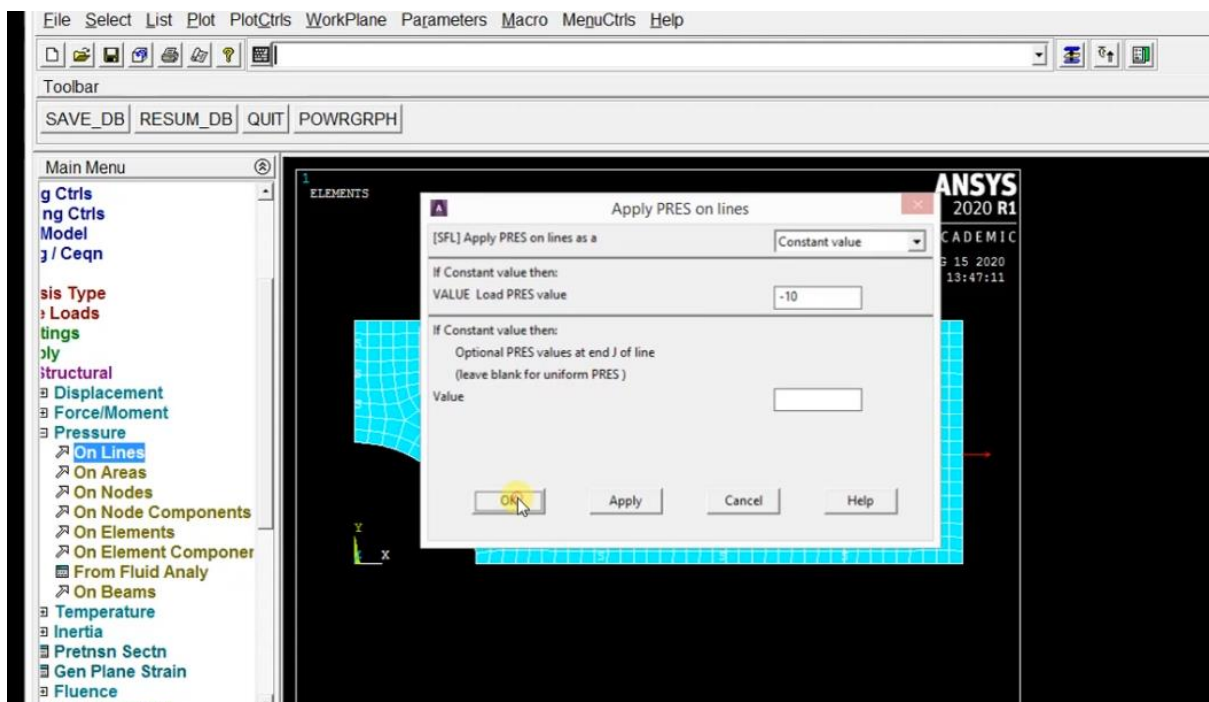
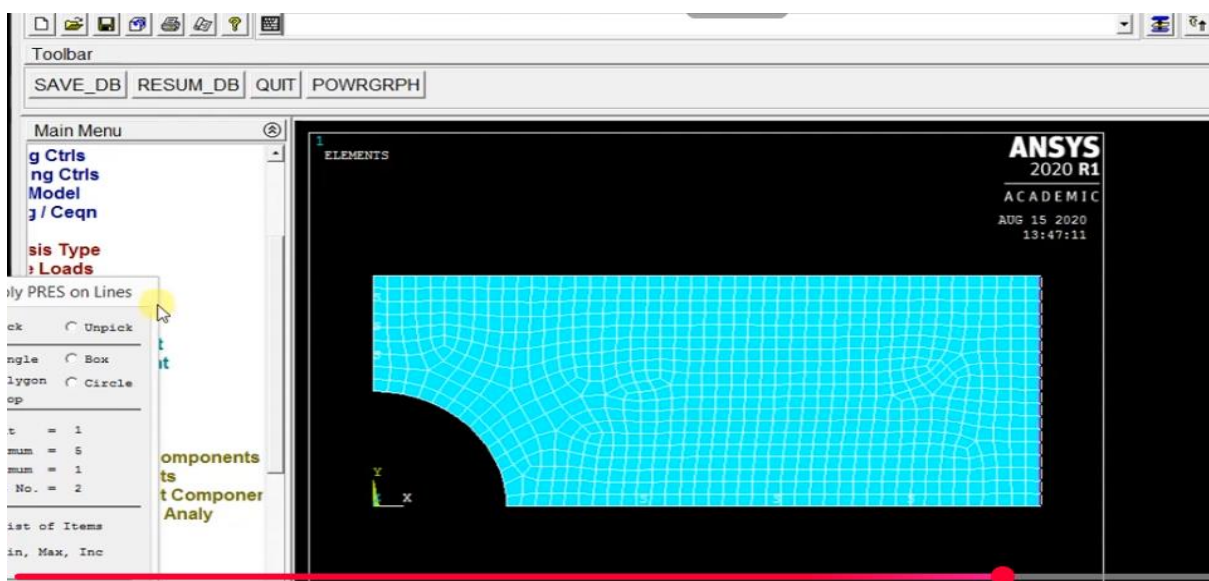
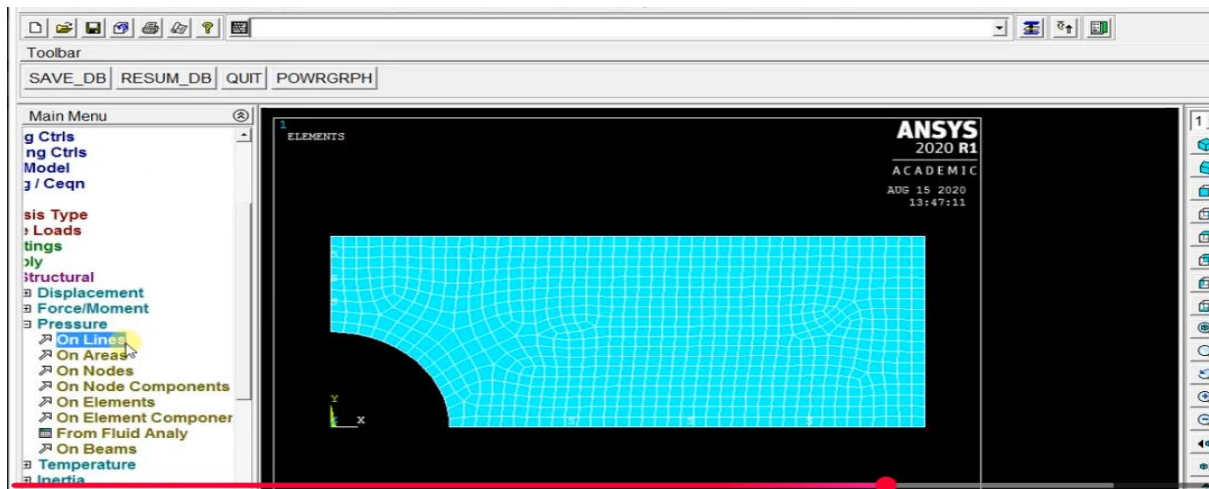
$$P = \text{Pressure} = 10 \text{ N/mm}^2$$

Reason :

In calculation *force* is used to calculate maximum stress, but

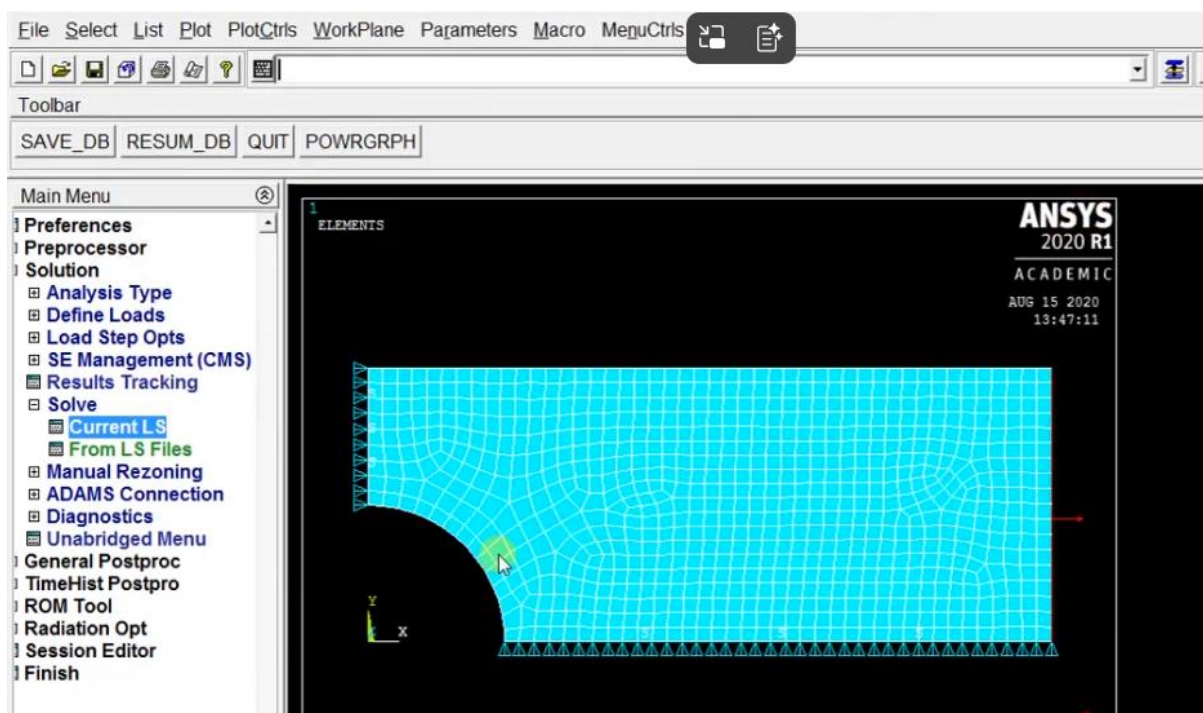
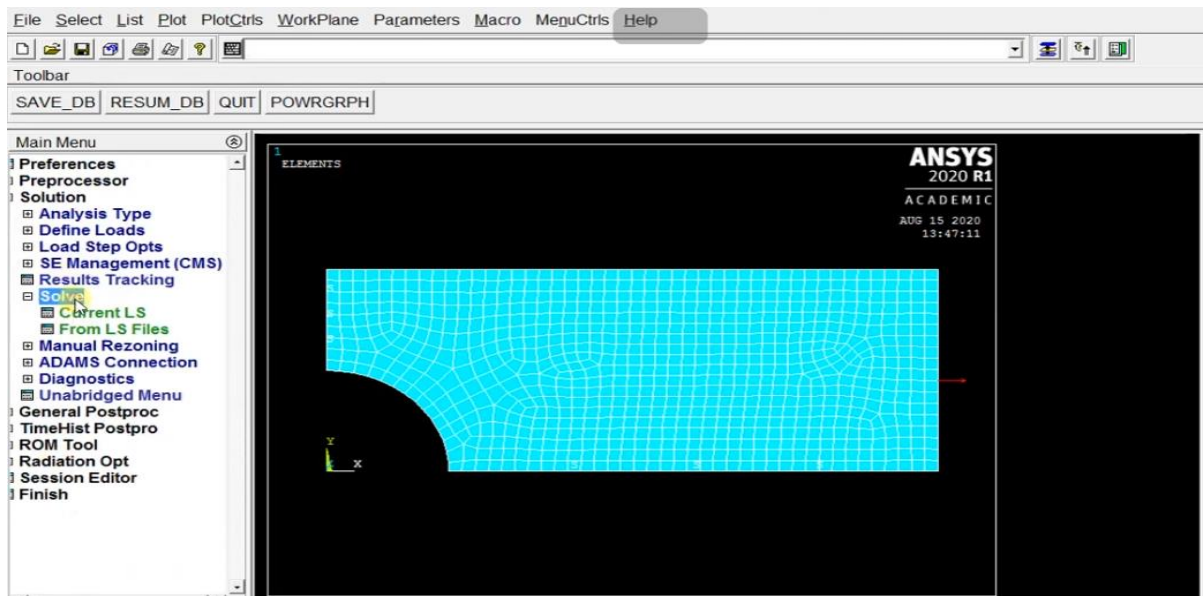
In ANSYS *pressure* is used to calculate maximum stress

Activi:
Go to S



Step 8 Solution

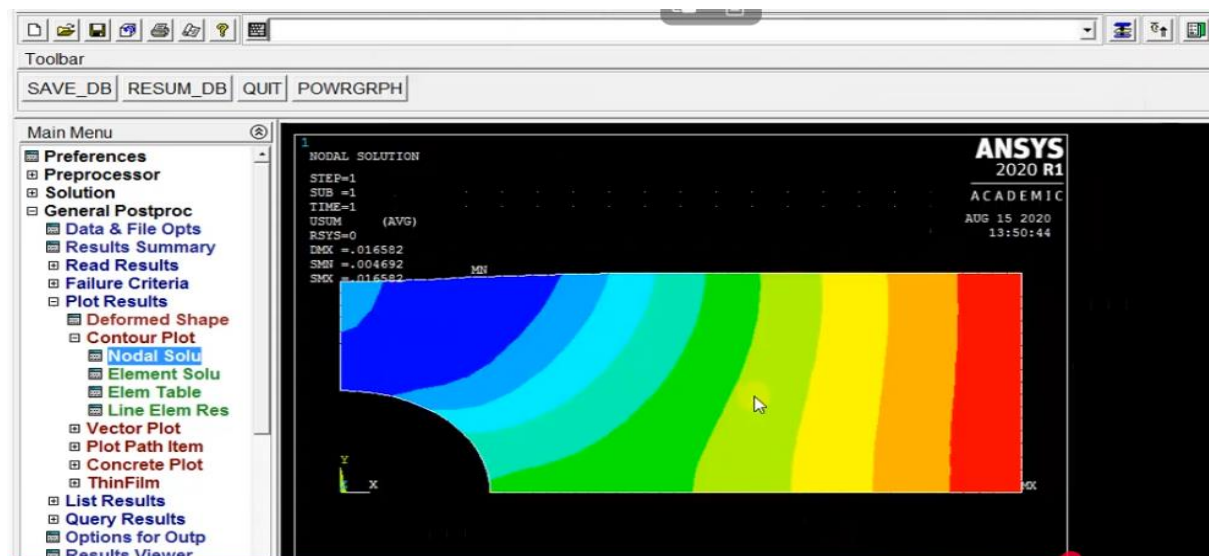
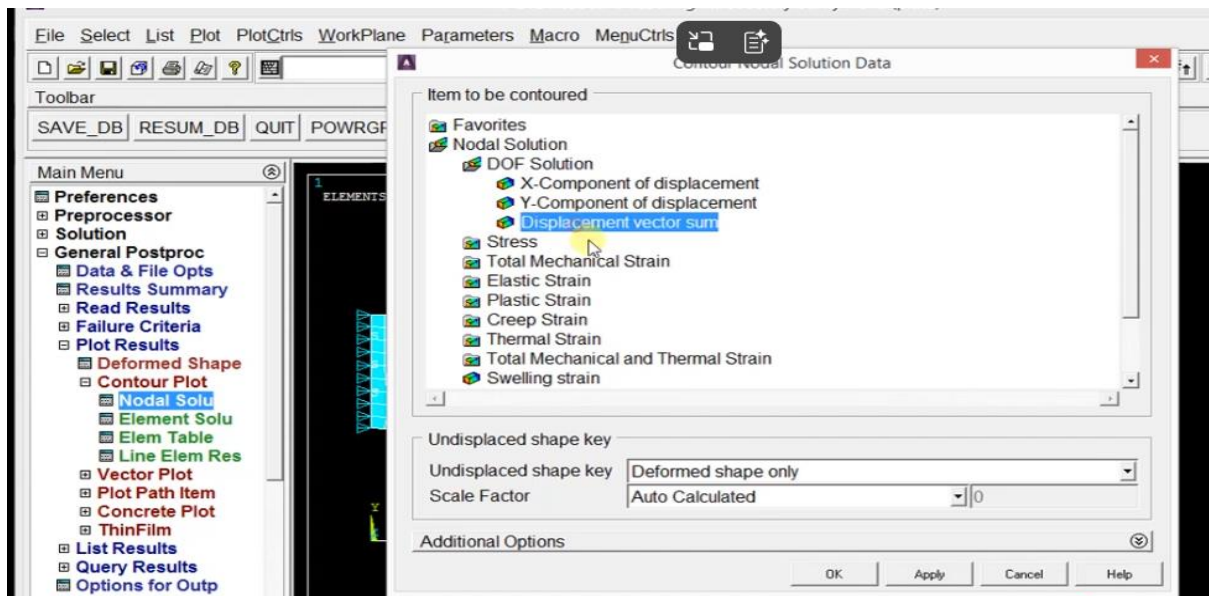
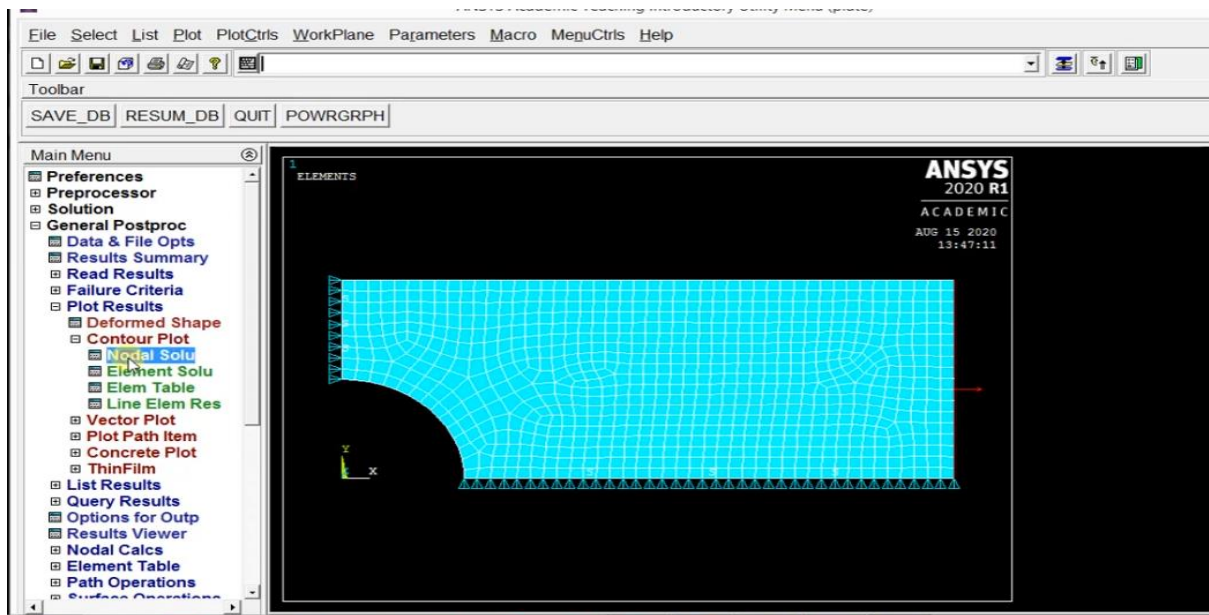
Solution – Solve – Current LS – OK



Step 9 General Results

General Post Process – Plot Results - Contour Plot – Nodal Solution – DOF

Solution – Displacement Vector Sum - OK

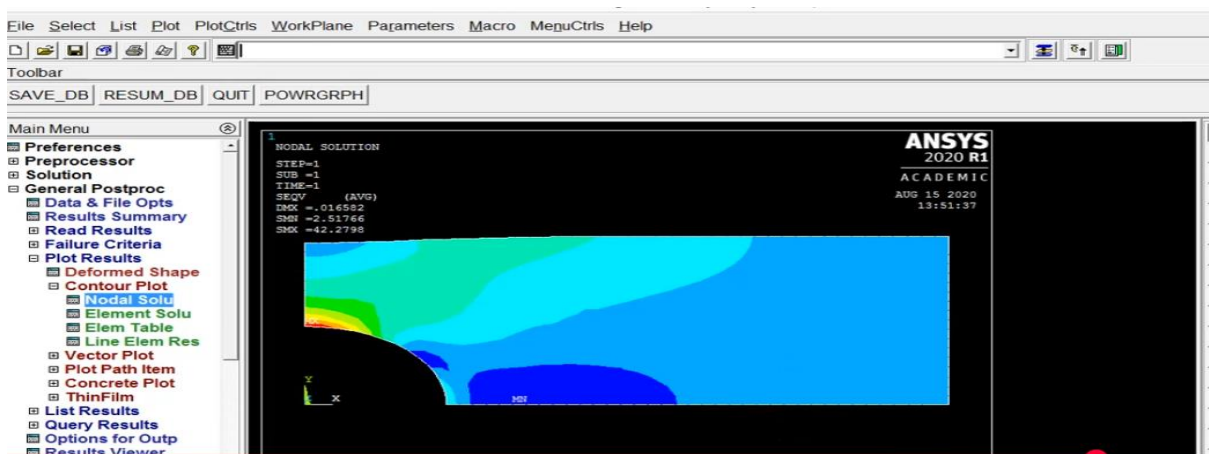
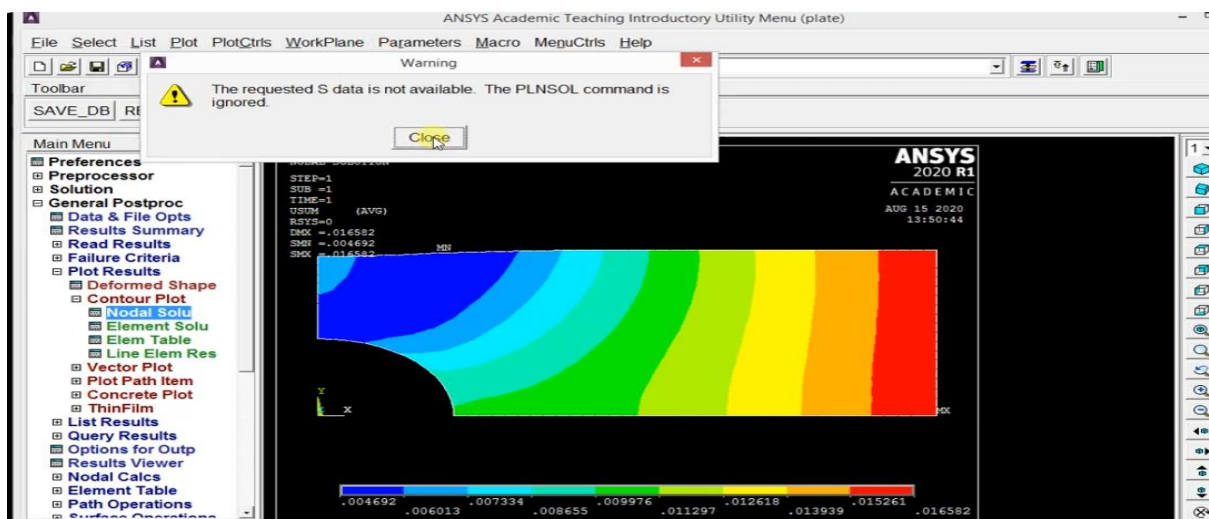
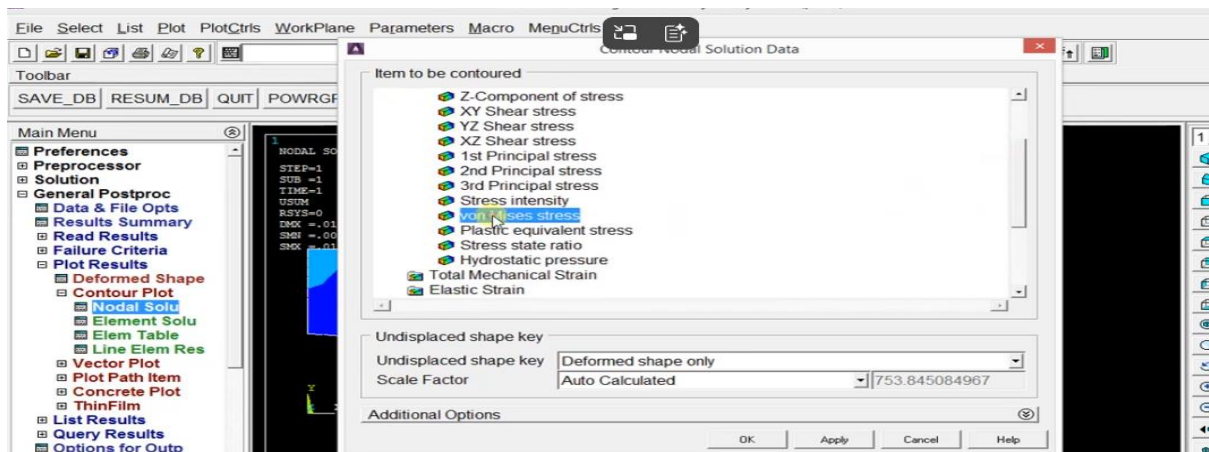


To Find Stresses

Apply Command – Set, last

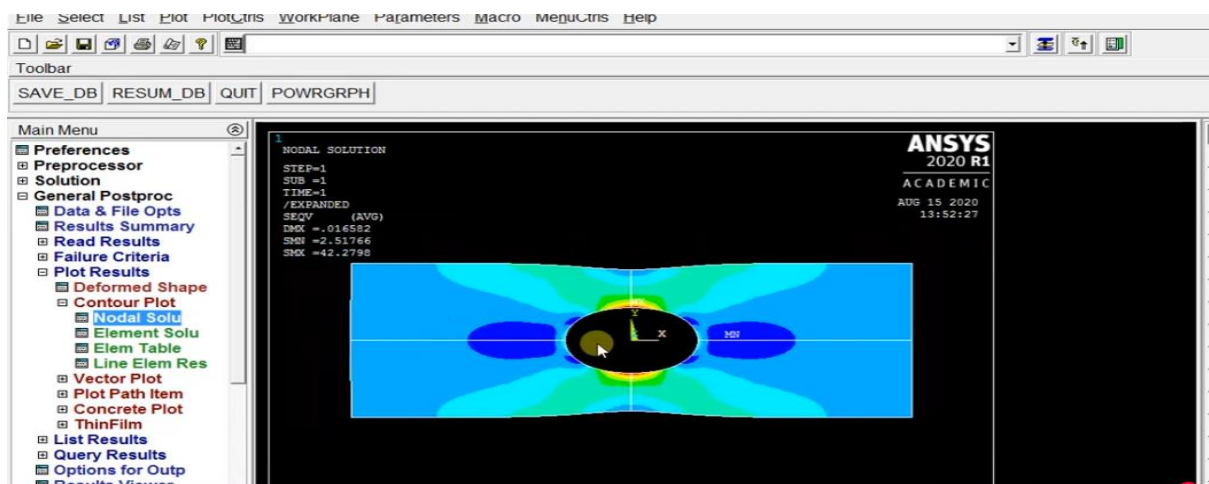
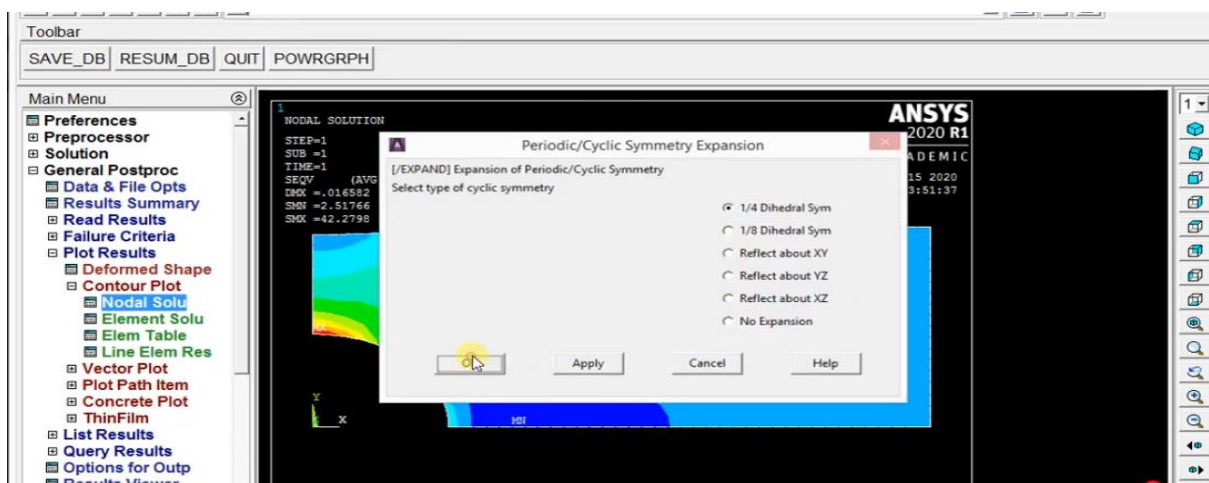
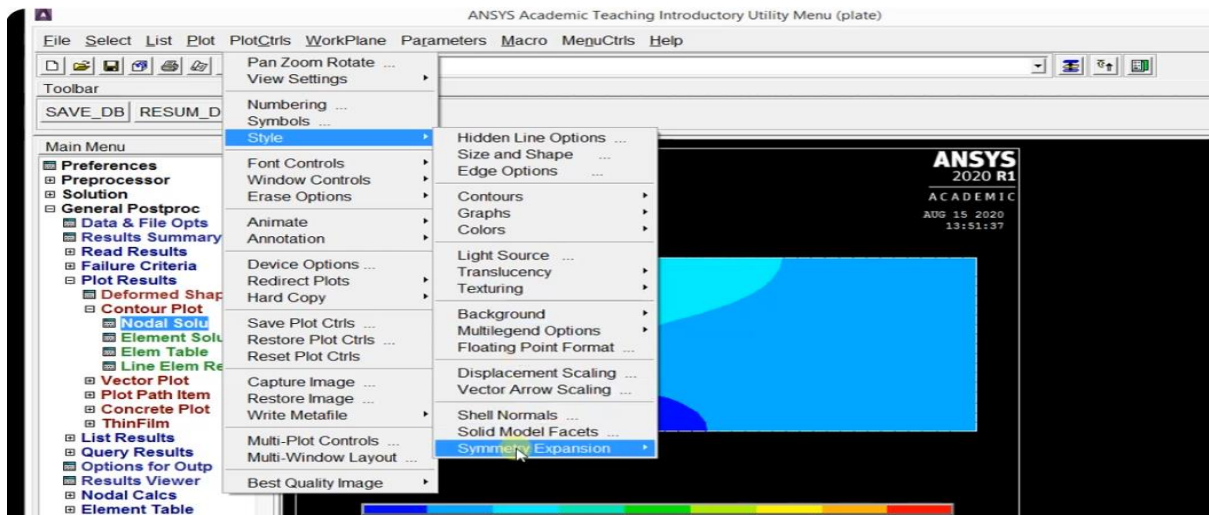
General Post Process – Plot Results Contour Plot – Nodal Solution – Stresses

– Von Mises Stresses - OK



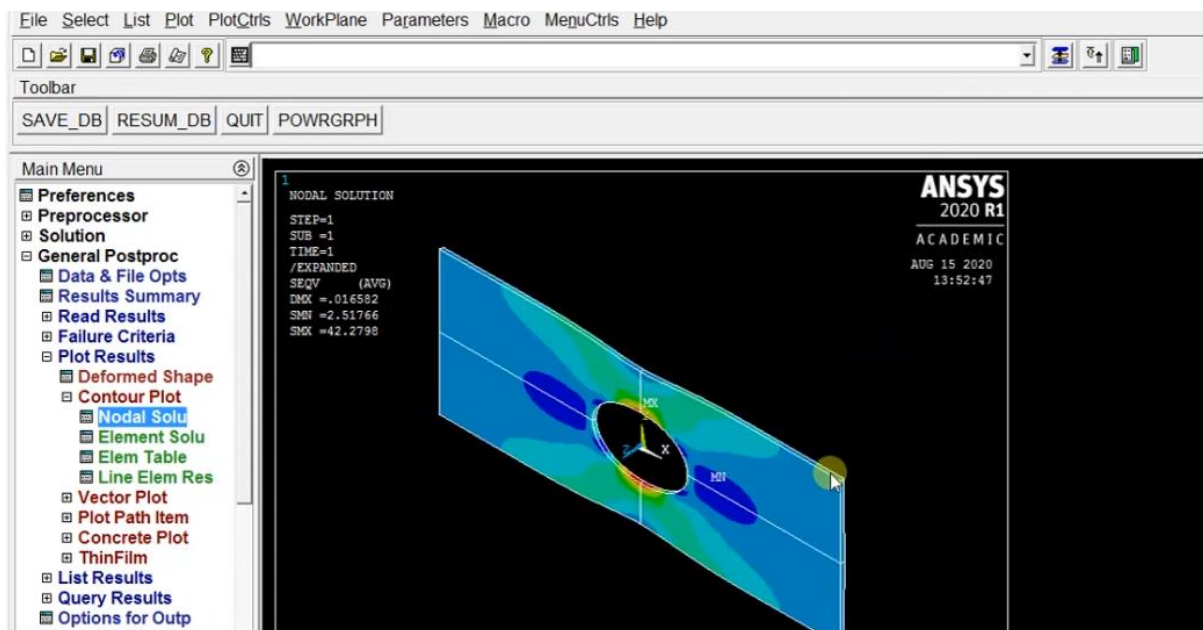
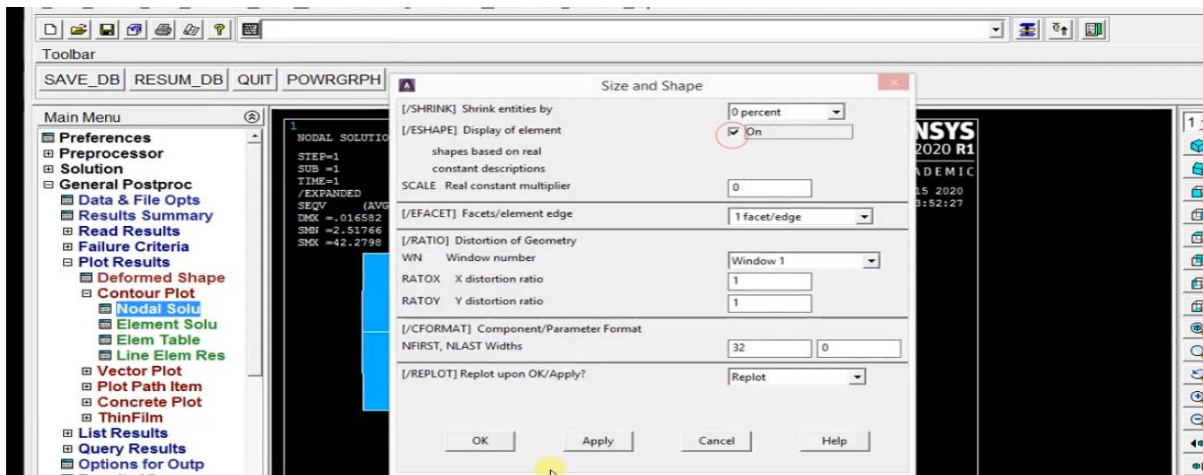
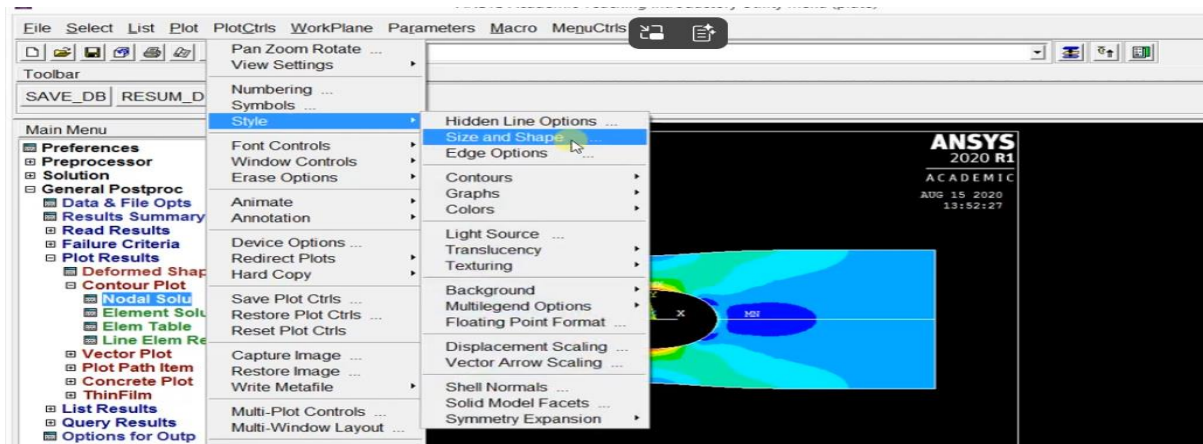
To See Actual Plate

Plot Controls – Style – Symmetry Expansion – Periodic Cycle – select $\frac{1}{4}$ symmetry - OK



To See Thickness of Plate

Plot Controls – Style – Size and Shape – Display of Element on – OK



To See Animation

Plot Control – Animate – Deformed Results – stresses – von mises stresses -

ok

